

# FRANCESCA MONTE

Marketing Data Analyst · Analytics Engineering · Digital Strategy

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Open to relocation: Amsterdam / Dublin

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## PROFILE

*I've always needed to know the reason behind the result. Not just what worked — why it worked, and what would happen if you changed one variable. That instinct pushed me from content strategy into data analytics: I couldn't accept making decisions I couldn't verify. Now I build the analysis that answers the questions most people don't think to ask.*

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## SELECTED PROJECTS

**SENTRY.mark — Automated Anomaly Detection & Alerting System** · BigQuery · Python · Power BI · Slack · GitHub Actions May 2025

*End-to-end pipeline detecting statistically significant deviations in GA4 marketing data with real-time alerts and a budget kill-switch.*

- Queries GA4 data from BigQuery; detects deviations via rolling Z-Score baselines (7-day window,  $\alpha=0.05$ , threshold  $Z>1.96$ )
- Three-tier architecture: Detection → Classification → Activation; severity bands MEDIUM / HIGH / CRITICAL with business impact mapping per anomaly type
- Kill-Switch simulator triggers Slack alerts and budget suspension recommendations only when anomaly type and  $Z\geq 3.0$  are simultaneously met — prevents false positives
- Validated via chaos engineering: 5 synthetic anomalies injected into real GA4 data — all 5 detected correctly
- Executive Power BI dashboard with Command Center + Deep Dive views; pipeline automated via GitHub Actions

**Stack:** Python · Pandas · SciPy · BigQuery · SQL · Power BI · Slack Webhooks · GitHub Actions

**Causal Impact of Ad Exposure on Conversion Rate** · Causal Inference · Python · Statistical Analysis Mar 2025

*Measuring the true incremental lift of advertising using Inverse Probability Weighting (IPW) — separating ad effectiveness from user selection bias.*

- Identified and corrected selection bias: users shown ads were systematically younger, more active, and more mobile — making naive conversion comparisons invalid
- Estimated propensity scores via Logistic Regression; applied IPW weights to create statistically balanced exposed vs. control groups
- Validated covariate balance using Standardised Mean Differences (Love Plot): all SMDs below 0.1 threshold after weighting
- Result: raw lift 6.05% vs. true causal lift 5.17% — 0.89pp overestimation bias attributable to user characteristics, not ad effectiveness

**Stack:** Python · Pandas · Scikit-learn · NumPy · Matplotlib · Seaborn

**Amsterdam Airbnb Market Analysis — ETL Pipeline & Dashboard** · Python · BigQuery · Power BI · Data Engineering May 2025

*End-to-end data pipeline: raw CSV ingestion → Python cleaning → BigQuery cloud storage → multi-page Power BI dashboard. Dataset: ~10K listings, 3.8M calendar rows, 500K reviews.*

- Built full ETL pipeline from Inside Airbnb raw data: cleaning and transformation in Python, cloud load into BigQuery (3 relational tables), Power BI data model with relationships
- Key finding: Centrum-Oost leads revenue potential (~€60K/listing/year); Bijlmer-Oost shows highest occupancy driven by supply constraint, not demand excess
- Price-quality analysis: above €500/night, price does not correlate with review scores — Amsterdam premium listings sell on location, not experience

**Stack:** Python · Pandas · Google BigQuery · Power BI · DAX

**Additional project — E-commerce Profitability Attribution (SQL · BigQuery):** identified hero product categories at 62% gross margin via 3-table JOIN analysis; HVC segmentation for Lookalike Audiences. Full portfolio →

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EXPERIENCE

- Freelance Marketing & Analytics Consultant — Independent

Jan 2026 – Present

  - Meta ad campaign audit, performance analysis and budget strategy for an emerging restaurant in a competitive urban market
  - Website performance review and SEO recommendations based on traffic and behavioural data
  - Projects SENTRY.mark and Causal Inference on Ad Spend originated directly from this consulting work
- Data Quality & AI Content Evaluator — RWS · Remote, project-based · (concurrent)

Sep 2025 – Jan 2026

  - Evaluated AI model outputs for Meta and Google ad systems against GDPR and data classification standards
  - Search quality evaluation: assessed SERP relevance against user intent using strict platform guidelines
- Content Strategist & Media Manager — Polymed Surgery S.r.l · Naples

Feb 2025 – Nov 2025

  - Grew Instagram following from ~60K to ~70K (+15%) in 9 months through data-informed editorial planning
  - Managed and analysed Meta ad campaigns: reach, follower growth, and active client acquisition
  - Full ownership of content strategy, editorial calendar, and paid campaign execution
- Marketing & Communications Intern — Asco Pompe S.r.l · Milan

Oct 2024 – Jan 2025

  - Brand strategy project spanning multiple subsidiaries under a single corporate umbrella
  - Stakeholder management across CEOs of different business units to build a unified brand identity
  - Competitive landscape analysis and brand positioning across the industrial sector
- Marketing & Social Media Intern — Royal Group S.r.l · Naples

Jan 2023 – Jan 2024

  - Supported digital campaign execution and monitored performance metrics across paid and organic channels

TECHNICAL SKILLS

**Analytics & Data** BigQuery · SQL (CTEs, JOINS, multi-table) · Python (Pandas, SciPy, Scikit-learn) · Power BI (DAX, Star Schema) · GA4 · Looker Studio

**Methods** Statistical Analysis · Causal Inference (IPW) · Anomaly Detection · ETL Pipeline Design · Experiment Analysis · KPI Reporting

**Marketing** Meta Ads · Campaign Analysis · Editorial Strategy · SEO · Brand Strategy · Content Strategy · GTM

**Automation** GitHub Actions · Slack API · n8n · Agentic AI Workflows · Git

**Compliance** GDPR · Data Classification

EDUCATION

- Master's — Strategic & Professional Communication · University of Pisa

Dec 2024
- Bachelor's — Languages & Cultures (Europe & Americas) · UniOr

Oct 2023
- Digital Marketing & Agentic AI Master · Start2Impact University

Expected Aug 2026

CERTIFICATIONS

- Google Data Analytics Professional Certificate · Google / Coursera

Feb 2026
- Google Analytics Certification · Google

Jan 2026
- Microsoft PL-300 — Power BI Data Analyst Associate · Microsoft

Expected May 2026

LANGUAGES

Italian (Native) · English (C2) · French (B2) · Portuguese (B2) · Dutch (A1 — actively learning)